

Human Patience and AI Frustration: How Repeated Chatbot Errors Relate to Users' Trust, Attention, and Intended Behavior

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Google Forms survey results

Abstract

AI Chatbots have been used a lot more in education, workplace, and problem-solving in daily life activities. Nevertheless, repeated errors might affect the way users behave when dealing with such systems. This cross-sectional, scenario-based survey examined reported frustration, trust, reading attention, and intended behavior after repeated AI chatbot errors. Altogether 204 submissions were collected. Exact-content deduplication retained 181 unique response patterns and removed 23 later duplicates. Frustration increased strongly across both scenarios: Scenario A rose from 2.93 after one incorrect answer to 5.77 after eight, and Scenario B rose from 3.34 early to 5.59 near the end. Participants had moderate expectations for AI correctness ($M = 4.14$) but lower overall trust ($M = 3.56$). Rebuilt inferential models supported H1, produced mixed evidence for H2, partially supported H3 for any reported attention change, and did not support H4: frustration did not predict the grouped switch-away outcome, $OR = 1.08$, 95% CI [0.84, 1.38], $p = .565$. The results describe hypothetical ratings and intended actions, not observed causal effects or actual chatbot behavior.

Introduction

AI chatbots are becoming a huge part of our daily digital experience and shapes the way we tackle issues and problems. Students use these tools to get assistance with their studies, generate ideas, and write. Even adults and employees use them to become more productive; and regular users turn to them for summarizations, workout routines, and other general suggestions. The fluency of chatbot responses leads to users regarding them as valuable resources regardless of

their lack of certainty about the validity of the provided answers. Thus, one encounters an important human-AI interaction issue. What to do when a seemingly helpful resource provides wrong answers over and over again. Previous work on large language models has revealed that fluent responses could be unjustified, misleading, and factually wrong (Bender et al., 2021; Huang et al., 2023).

When it comes to AI chatbots making a mistake, a single error may be annoying but is not a huge problem for most users and is usually swept away. On the other hand, repetition of errors is different. A repetition of errors can affect how the whole process unfolds and what decision the user makes: to continue relying on the bot, to ask again or to check the answer, to use some other software or to give up on the task altogether. In this scenario-based survey, the repetition of mistakes made by AI is viewed as a problem of frustration as well as a problem of behavior decisions.

The errors made by AI systems or chatbots are very different in nature compared to those committed by a typical conventional software. While a button or page freeze occurs through a detectable and predictable error pattern, an AI assistant such as a chatbot may correctly answer one question and then provide an incorrect response to the very next request posed by the user. This unpredictability makes it difficult to gauge the level of trust in the system since users will have no clear idea when the AI system will provide accurate information. What is more, the core concern lies not only in whether the chatbot provides an inaccurate response, but whether the user can predict that it will make a mistake.

This research paper mainly concentrates on regular users of AI and tries to determine how the constant errors of chatbots influence users' frustration, trust, attention spans, and behavioral intentions, such as entirely giving up or switching to another tool. The primary thesis supported

by the results of the survey is that the errors of AI cannot be reduced only to creating frustration. These repeated mistakes from AI chatbots could prompt the user to read its responses more carefully, rephrase prompts, or check information using other alternatives.

Research Question and Hypotheses

Research Question: What is the relationship between repeated AI chatbot mistakes and reported user frustration, trust, reading attention, and behavior such as rephrasing, switching tools, or giving up?

- H1: As the number of AI chatbot mistakes increases, the frustration levels of users will also increase.
- H2: Inconsistent AI mistakes will induce similar or greater frustration than consistently wrong mistakes since users cannot easily determine when the AI can be trusted.
- H3: Greater frustration will be correlated with self-reports of changes in reading attention.
- H4: Greater frustration will be correlated with intentions to drop out or switch away, which can be assessed by user reports of quitting, switching to a different AI, or switching to Google.

Literature Review

Human-Computer Frustration and Repeated Errors

From research conducted into human-computer interaction, it is clear that frustration occurs when technology gets in the way of user objectives, wastes time, or causes the user to perform additional tasks for recovery purposes. As stated by Ceaparu et al. (2004) in their study into end-user frustration, computer problems are not simply technological in nature but emotional as well because computer problems cause people to waste time and effort when they try to achieve an objective. Similarly, Lazar et al. (2006) state that computer frustration may have tangible effects on users since computer frustrations may include blocking or causing users to repeat tasks. This is relevant to chatbot design since the incorrect response from a chatbot will not just indicate one erroneous output.

These previous studies that examined HCI explained why repetitive errors of AI can be more harmful than one error alone. While, in case of a software error, the user will notice that there is an error made by the software and perhaps know why it occurred. In case of a mistake made by

AI during a conversation with the bot, the error will appear within the sentence which may look correct. It will be really hard to find the mistake and can lead to an unpleasant result. People can ask help with an issue about which they know nothing from AI. But since the bot replied incorrectly, people will use the reply regardless of its response. Therefore, the frustration during a conversation with the bot will not only result from the mistake but trying to establish whether it is useful or not.

Trust in Automation and Trust Calibration

"Trust" is one of the most important concepts that connect the areas of automation research and AI chatbot implementation. According to Lee & See (2004), trust in automation becomes critical for users because they have to use automated tools in those cases when it is useful and not use them in those cases when their actions are incorrect. It means that the purpose is not the maximal level of trust, but the appropriate level of trust. Problems of misuse, disuse, and abuse of automation were introduced by Parasuraman & Riley (1997). In other words, there is an opportunity for excessive reliance on automation, rejection of automation, and misuse of automation. For AI chatbot implementation, it means that the user's attitude towards errors will depend on the ability of the chatbot to help in calibration of trust.

The current research also relates to algorithm aversion. Dietvorst et al. (2015) have established that humans tend to reject algorithms' recommendations after making a mistake and regardless of how well the algorithm may work on average. It may prove helpful for analyzing chatbot frustration because the user will not just evaluate the quality of the interaction based on the overall average value of the algorithm. On the contrary, the error will greatly diminish the subsequent trust in the algorithm. In this case, the problem is reflected in the question about trust reduction and the first action question because several respondents started using Google or another website after receiving several incorrect responses from the chatbots.

Cognitive Load and Verification Effort

The "Cognitive Load Theory" provides us another framework to interpret how chatbot errors may frustrate users. According to Sweller (1988), there are limitations on the working memory capacity that humans possess during problem-solving and that learning or performing the task will be adversely affected by any additional cognitive load which is unnecessary. Mayer (2005) further pointed out the need to make information designs such that the user's attention is devoted to only the primary task. In terms of artificial intelligence chatbots, the additional cognitive load stems from the fact that the user will have to evaluate the answer and verify its correctness.

This concept as described, directly relates and connects to the theory of the verification shift as proposed in this research paper. The user needs to concentrate only on the task at hand if the

provided AI-based response is trustworthy. If it is potentially unreliable, the user should concentrate on both the task and the reliability of the tool. This additional load may help explain why respondents in the current survey reported reading the text more carefully after imagined mistakes. Although it may seem like an increased level of attention, increased verification due to decreased trust is another possible interpretation of it.

AI Reliability, Hallucination, and Fluent Wrong Answers

The latest AI chatbots are based on language models capable of generating fluent texts similar to human writing. Yet, studies on language models have found several times that fluent language does not necessarily mean accurate facts. According to Bender et al., (2021), large language models can create fluent language but lack grounded knowledge. As a result, people may take this information as something more reliable than it is supposed to be. The latest studies on large language models' hallucinations show how such models may generate plausible but non-factual information, particularly, in case an answer requires factual grounding or goes beyond the area where it is reliable (Huang et al., 2023; Ji et al., 2023).

Hallucination research is relevant for the current research paper since it describes the reasons why errors made by AI are hard for users to deal with. A response made by a chatbot can be wrong and at the same time be confident, polite, and fluent, so it's hard for users to differentiate on what is right or wrong. Therefore, there is an altered user experience compared to searching information online via engines that offer sources or various links. In case when a chatbot gives a single response, users may have to find sources themselves. It all fits into the topic of verification behaviors that users tend to display after errors are made.

Crowdsourcing, Online Survey Behavior, and the Current Gap

Due to the fact that this paper used an online Google Form survey, it is also necessary to mention literature on online samples and crowdsourcing participation. Kittur et al. (2008) demonstrated that online labor markets could be used for user studies, but it also means that researchers should pay more attention to participant attention and data quality. Gadiraju et al. (2015) found out that online studies could include low-effort or malicious participation, which means that data filtering is crucial. It is necessary to discuss this point because this paper used a convenience sample and found duplicate content among participants later on. To remove duplicate content among participants, google sheet tools were used.

Ultimately, the literature shows that the experience of repetitive errors of an AI chatbot is related to several topics of research in psychology and HCI including frustration with technology, trust in automation, cognitive load, algorithm aversion, and reliability of AI. Current research explains the nature of frustration with technology errors, the need for proper calibration of trust, and the

fact that fluency in answers from an AI program does not mean that it provides reliable results. Still, there is limited information regarding the ways in which users perceive their actions and behaviors in case of repetitive mistakes of a chatbot.

Method

Design

The research design applied to this study was cross-sectional, scenario-based. Participants took one Google Forms survey where they provided their information regarding AI usage, their trust towards the AI chatbots, their frustration when faced with repeated errors, change in attention and potential actions they would take after experiencing repeated wrong answers. The fact that all participants went through Scenario A and Scenario B reveals that the study can be considered as a within-subjects scenario comparison study embedded in a cross-sectional survey, not a live chatbot experiment where real life experiments would take place.

Participants

There were 204 submitted surveys. During the data collection process, 23 duplicates based on content were found, and there were 181 unique patterns of responses left for cleaned descriptive analysis. To clear out these duplicate responses, we used built-in google sheet tools. This sample was a non-randomly recruited convenience sample of online and school/social-network respondents who had experience with AI chatbots.

Table 1

Participant and sample summary

Measure	Result
Submitted responses	204
Cleaned unique response patterns	181
Multiple-times-a-day AI users	62 (34.3%)
Intermediate AI users	97 (53.6%)
Most common age group	25-34

Materials

The survey was titled as “AI Interaction Experience Survey” and was sent out via Google Forms. In order to share the survey and to gather responses, the google form was shared in media such as Reddit, SurveyJunkie, and to close members. It contained AI interaction questions, trust and accuracy expectations, two error repetition scenarios, attention and behavior questions, a modified frustration scale of five items, general reliability and consistency expectations, a voluntary open-ended response, and demographic questions.

Scenario Conditions

In Scenario A, there was a scenario involving consistent AI failure. The participants envisioned themselves interacting with an AI in search of 10 facts, receiving incorrect responses at each step of the way. Frustration ratings for one incorrect answer, three incorrect answers, five incorrect answers, and eight incorrect answers were asked for.

Scenario B involved inconsistent AI interaction. The participants envisioned themselves interacting with an AI that gave correct responses to certain questions and wrong responses to others, leaving them unsure as to which answers to believe in. Frustration ratings were collected early on in the process, halfway, and close to the end.

Measures

AI usage at baseline was assessed by usage of the chatbot platform, usage frequency, and experience level with AI. Trust was assessed by factual accuracy, general trust, and trust reduction due to an incorrect response, on a 1-7 scale. Frustration trajectory was assessed through the Scenario A grid and Scenario B grid. Attention was assessed by reading behaviors and a threshold for shifting attention. Intentions to drop out or switch away was assessed with a first action question following wrong responses. A main frustration scale was developed based on five items regarding giving up, time use, mental drain, confidence, and mental effort. The

positive items were reversed prior to averaging. Reliability and consistency were also measured on a 1-7 scale.

Procedure

Participants opened the Google form, reviewed the instructions and questions, completed questions that assessed for AI chatbot usage, completed the trust and expected accuracy measures, followed by the two scenarios with wrong responses, followed by the attention and behavior measure, then completed the adapted frustration scale, reliability and consistency measures, and finished with demographic questions.

Data Cleaning and Analysis Plan

This research used descriptive statistics and rebuilt inferential models. Categorical responses were summarized with counts and percentages; 1-7 ratings were summarized with means and standard deviations. Exact-content deduplication retained the first occurrence of each response pattern and removed 23 later submissions that matched an earlier submission across all response fields except timestamp, leaving 181 patterns. H1 used Friedman tests and Holm-adjusted paired Wilcoxon tests. H2 used an exact binomial forced-choice comparison and a paired final-checkpoint sensitivity test. H3 and H4 used binary logistic regression with odds ratios and 95% confidence intervals, plus nonparametric sensitivity analyses. Switch-away combined Google/search, a different AI, and giving up; rephrasing and retrying were coded as stay-and-repair. The 37 optional written responses are presented as illustrative comments rather than a formal thematic analysis. The models describe associations in a nonprobability scenario survey and do not establish causal effects.

Results

Participant Characteristics

In total, after cleaning the dataset, there were 181 unique response-content patterns. After reviewing our

data, we can see that the majority of participants used AI technologies regularly. From our data, 62 participants (34.3%) used AI technologies several times per day, and 51 participants (28.2%) a few times per week. Most participants regarded themselves as intermediate users (97 participants, 53.6%), then beginner users (44 participants, 24.3%), and advanced users (40 participants, 22.1%). ChatGPT Free was the most popular chatbot platform among participants (97 participants, 53.6%), then Gemini (38 participants, 21.0%), and Claude (37 participants, 20.4%).

Table 2
Chatbot platforms selected by respondents

Category	n	%
ChatGPT (Free)	97	53.6
Gemini	38	21.0
Claude	37	20.4
ChatGPT (Plus)	30	16.6
Other	25	13.8
Copilot	25	13.8

Baseline Trust and Accuracy Expectations

Moderately accurate responses were expected, meaning users somewhat trusted AI chatbots accuracy, ($M = 4.14$, $SD = 1.56$) although overall trust level was lower ($M = 3.56$, $SD = 1.62$). The drop in trust level after an incorrect response was found to be high ($M = 4.50$, $SD = 1.77$). Overall rating of general reliability was lower ($M = 3.39$, $SD = 1.47$), but the rating of consistency or predictability was higher ($M = 4.06$, $SD = 1.34$).

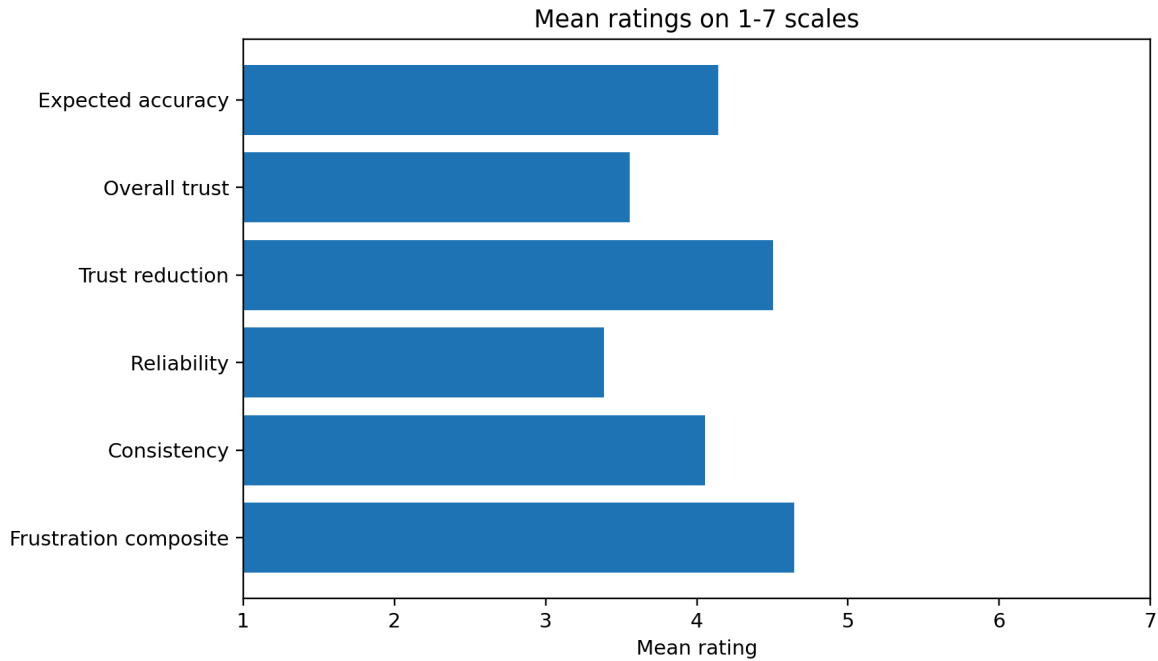


Figure 1. Mean ratings for trust, reliability, consistency, and frustration-related measures.

Frustration Increased Across Repeated Errors

In both scenarios, frustration levels were seen to increase as error rates accumulated. In Scenario A, the average level of frustration increased from 2.93 (SD = 1.72) when the first wrong answer occurred to 5.77 (SD = 1.68) after eight wrong answers. In Scenario B, the average level of frustration increased from 3.34 (SD = 1.58) early in the interaction to 5.59 (SD = 1.52) later on in the interaction.

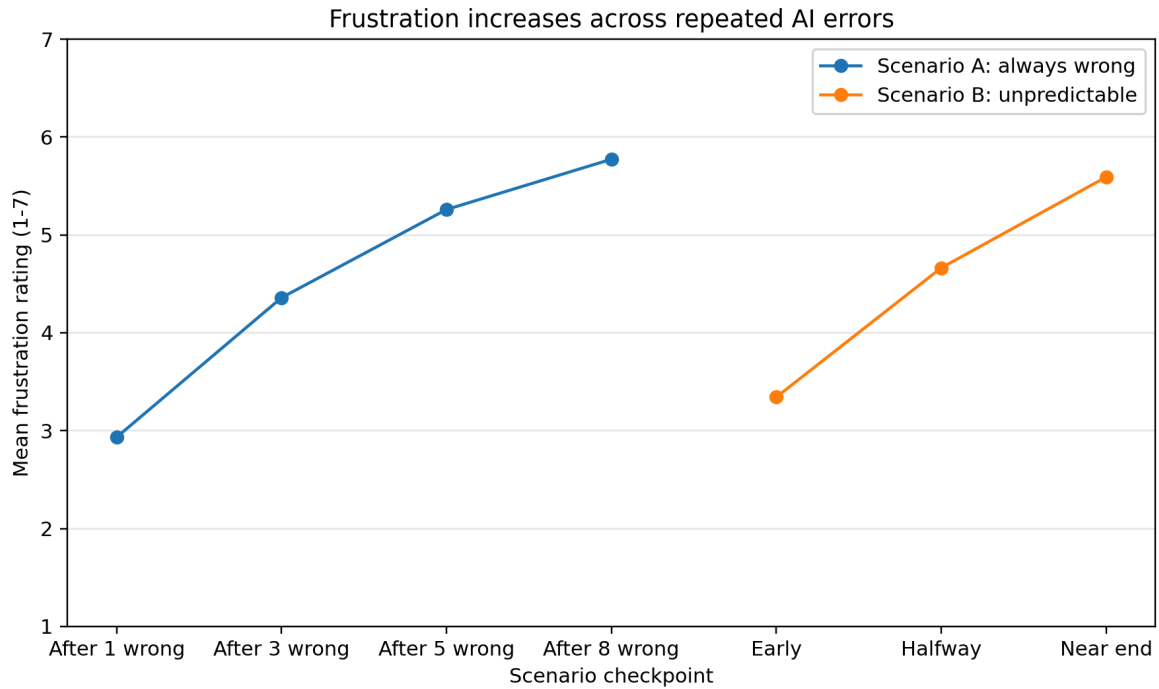


Figure 2. Mean frustration ratings across repeated-error checkpoints in Scenario A and Scenario B.

Table 3
Mean frustration ratings by scenario checkpoint

Scenario	Checkpoint 1	Checkpoint 2	Checkpoint 3	Final checkpoint
A: always wrong	2.93	4.36	5.26	5.77
B: unpredictable	3.34	4.66	5.59	N/A

Consistent Versus Inconsistent Failure

When participants were asked about their preference regarding types of failure that frustrated them more, the majority of 75 respondents (41.4%) said “both equally”. The number of participants who chose Scenario B (unpredictable error) was 60 (33.1%), while those who chose Scenario A (always-wrong errors) were 35 (19.3%). Only 11 (6.1%) participants chose neither.

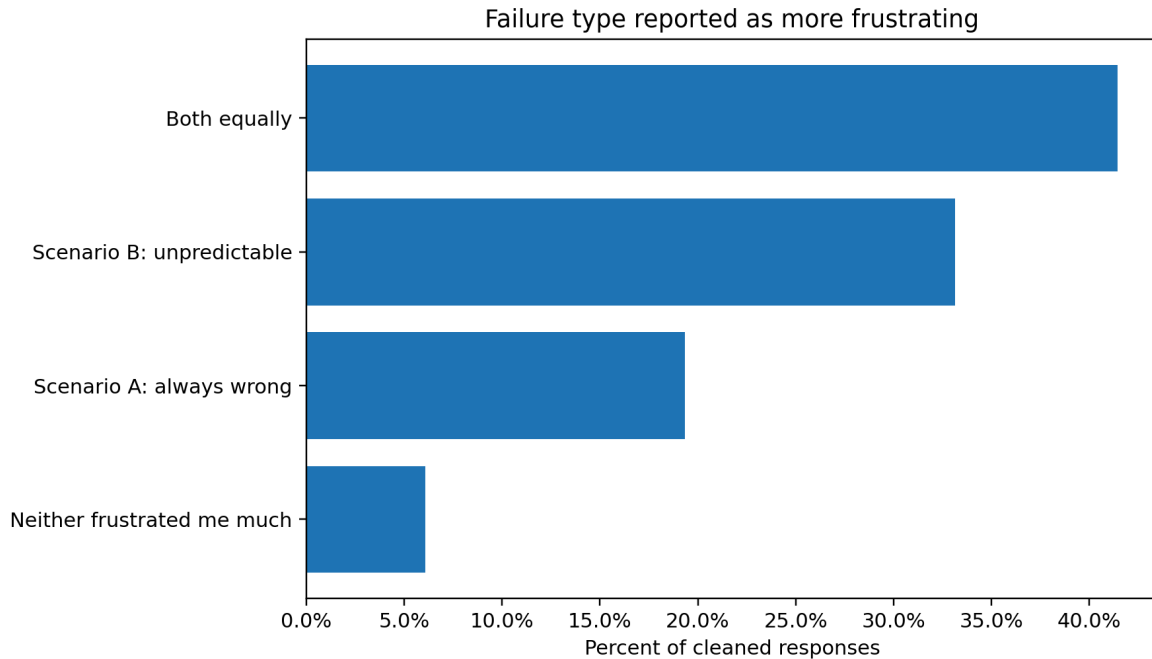


Figure 3. Failure type reported as more frustrating overall.

Attention Shift After AI Mistakes

The most popular choice that participants made was that repeating mistakes made them pay closer attention to reading AI responses (106, 58.6%). Fewer people reported skimming AI responses much less carefully (29, 16.0%), slightly less carefully (19, 10.5%), or not changing their way of interaction at all (27, 14.9%). Participants changed their way of interacting very fast: 58 participants (32.0%) changed it after one mistake, and 61 (33.7%) participants after two to three mistakes.

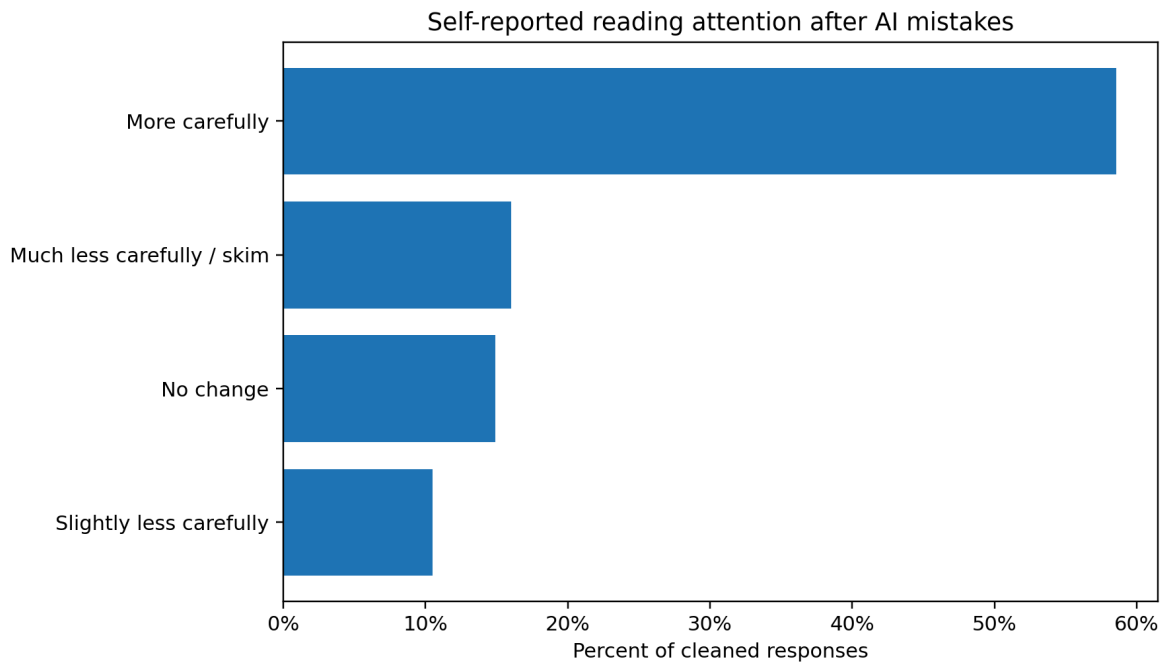


Figure 4. Self-reported reading-attention changes after AI mistakes.

First Action After Repeated Wrong Answers

The most popular reaction to getting repeated wrong answers is trying Google or any other search engine (71, 39.2%), closely followed by rephrasing or simplifying the question (69, 38.1%). Fewer participants tried switching to a different AI (21, 11.6%), giving up (16, 8.8%), or asking the same question again (4, 2.2%). Grouping switch away behaviors, such as switching to Google, switching to a different AI, and giving up, there are 108 respondents (59.7%) who demonstrated it.

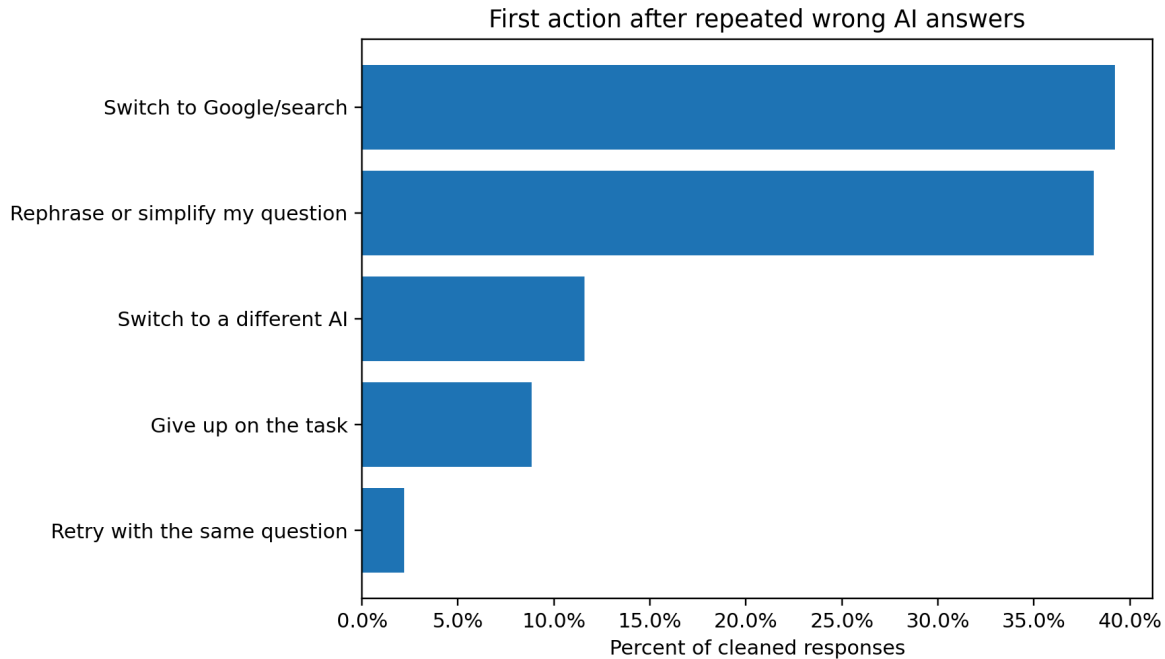


Figure 5. First action respondents reported taking after repeated wrong AI answers.

Frustration Composite

Frustration composite consisting of five items after reverse-coding positively oriented items on the use of time effectively and confidence in the AI has a mean of 4.65 (SD = 1.19). Cronbach's alpha for this five-item scale was .69, indicating marginal internal consistency for an exploratory, multidimensional composite.

Illustrative Comments From Written Responses

The optional written-response question was "What typically causes the participant to become frustrated with mistakes made by the AI chatbot?" In the Google Form summary, there were 37 optional written responses. The comments were reviewed for recurring topics to provide additional context for the quantitative findings. Because the supplied materials did not include a formal codebook, coder information, reflexivity statement, or documented theme-development procedure, the comments and quotations below are presented as illustrative rather than as a formal thematic analysis.

Table 4
Illustrative topics and participant quotes

Theme	Illustrative quote	Connection to findings
Verification and citation	<i>"If I can easily fact check. A good ai puts proper citations to where I can read more. And I often do read more."</i>	This finding / response from the user connects to the verification shift argument. Some users will check their sources before stopping due to errors.
Low expectations (low trust)	<i>"I generally don't get frustrated as I go in assuming that the answer will be wrong or incomplete..."</i>	Showcases the role that expectations play in frustration. Users who have a predisposition against artificial intelligence will experience less frustration due to errors.
Cost in time and effort	<i>"Ofentimes, the time saved on what it does right is not made up for by the time lost to fixing its mistakes."</i>	Links AI failures to the feeling of having one's efforts wasted, especially when users have to become human validators of AI work.
Repetitive failures and being unable to correct itself	<i>"When it doesn't acknowledge its mistakes or change its approach after being called out."</i>	Explains why repeated failures might lead to greater frustration: users expect that the system will learn from its mistakes and improve.
Knowing reason for problem reduces frustration	<i>"if it's a mistake due to a known and communicated limitation it frustrates me less..."</i>	Suggests that transparency regarding limitations can help in reducing frustration because failure becomes more understandable.
Imposed use and restrictions on access	<i>"It frustrates me the most when I am forced into it for customer service."</i>	Demonstrates that frustration is not only about inaccuracy but is enhanced when users feel that there is no other way for them.

To summarize this, the written responses provided additional insight into the patterns that were observed in the statistics. It seemed like frustration was often caused by wasting time, repeated mistakes, unreliable information sources, and the need for the user to verify information.

Statistical Analysis and Rebuilt Models

Cleaning rule and analysis sample

The rebuilt analyses used the 181 retained response-content patterns. Starting from 204 submissions, the first occurrence of each exact response pattern was retained and 23 later submissions that matched an earlier submission across every response field except timestamp were removed. This procedure identified repeated content, not verified unique individuals. No additional cases were excluded for straightlining or nonmonotonic ratings because no prespecified careless-response rule was supplied.

Data-quality check	Result	Analysis treatment
Raw submissions	204	Source rows
Retained response-content patterns	181	Primary analysis sample
Later exact duplicates	23 in 5 groups	Removed by stated rule
Missing core scale cells	0	No imputation
Invalid 1-7 values	0	No value exclusions
Scenario straightliners	17	Flagged, not excluded

Model specifications

H1 was tested with Friedman tests for repeated ordinal ratings, followed by paired Wilcoxon signed-rank tests using the Pratt treatment of zero differences and Holm adjustment. Kendall's W was reported as an omnibus effect size. H2 was evaluated with both the forced-choice comparison and a paired final-checkpoint sensitivity analysis. H3 and H4 were evaluated with binary logistic regression and reported as odds ratios with 95% confidence intervals. Spearman and Kruskal-Wallis tests were used for directional and category-level sensitivity analyses. These models describe associations within a nonprobability scenario survey; they do not establish causal effects or population prevalence.

The five-item frustration composite reverse-coded the positively oriented time-use and AI-confidence items before averaging. Because the scale combines frustration, effort, confidence, time appraisal, and giving-up intention, scenario-frustration measures were included as sensitivity predictors.

Inferential Results for H1 and H2

H1: Repeated errors increased frustration

Scenario A showed a strong checkpoint effect, Friedman chi-square(3) = 374.60, $p < .001$, Kendall's $W = 0.690$. Scenario B also showed a strong effect, Friedman chi-square(2) = 275.12, $p < .001$, $W = 0.760$. All successive paired contrasts remained below $p < .001$ after Holm adjustment. H1 was supported for the hypothetical repeated-rating trajectories.

Frustration rises across both hypothetical error sequences

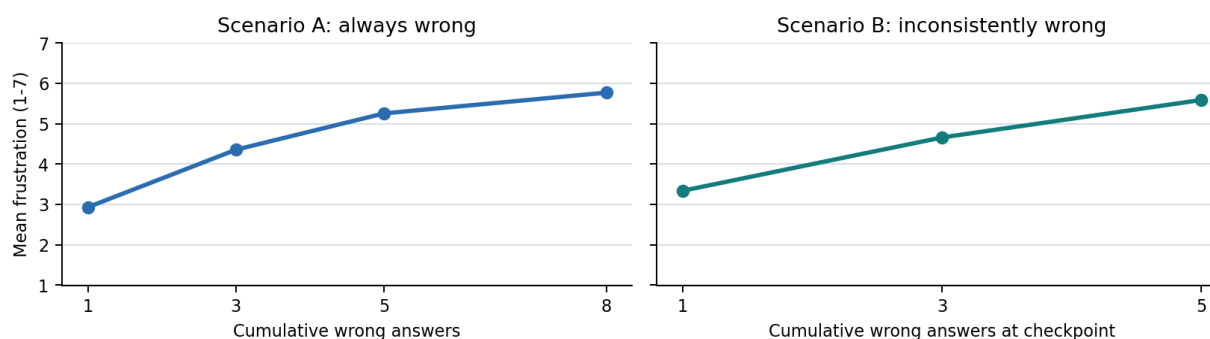


Figure 6. Rebuilt mean-frustration trajectories for the 181 retained response patterns.

Successive contrast	Mean difference	Holm p	Cohen dz
A: after 3 - after 1 wrong	1.425	< .001	1.209
A: after 5 - after 3 wrong	0.901	< .001	0.932
A: after 8 - after 5 wrong	0.514	< .001	0.616
B: halfway - early	1.320	< .001	1.207
B: near end - halfway	0.928	< .001	0.997

H2: Evidence was mixed

Among respondents who selected only Scenario A or Scenario B, 60 selected Scenario B and 35 selected Scenario A; the exact binomial p value was .013. However, paired final-checkpoint ratings were slightly lower for Scenario B than Scenario A, mean difference $B - A = -0.182$, Wilcoxon $p = .0095$, $dz = -0.174$. Scenario B was higher for 28 records, equal for 101, and lower for 52. Therefore, the forced-choice and paired-rating operationalizations did not converge, and H2 received mixed evidence.

Inferential Results for H3

H3: Frustration and self-reported attention change

Any attention change was coded as any response other than "no change" (154 of 181 patterns). The five-item composite did not reach the conventional .05 threshold, whereas the scenario-frustration measures predicted whether respondents selected any attention-change category.

Predictor	OR / point	95% CI	p	McFadden R2
frustration composite	1.389	[0.974, 1.980]	0.0694	0.022
A8	1.913	[1.495, 2.446]	0.0000	0.211
B5of10	1.681	[1.302, 2.170]	0.0001	0.111
terminal scenario mean	1.917	[1.470, 2.500]	0.0000	0.175
all scenario mean	1.767	[1.292, 2.415]	0.0004	0.091

Direction of the attention response

Coding the attention categories from 1 (much less carefully) to 4 (more carefully), composite frustration was negatively associated with direction, Spearman rho = -0.228, $p = 0.002$. The composite mean was 5.579 in the much-less-carefully/skimming group and 4.491 in the more-carefully group. The four category distributions differed, Kruskal-Wallis $H(3) = 21.53$, $p < .001$. H3 was therefore partially supported for any reported attention change, but the data did not establish a directional shift toward more careful verification.

The attention item was asymmetric: it offered one broad more-carefully category, two levels of less careful reading, and no slightly/much-more distinction. It also asked respondents what happened as AI made more mistakes. The category count of 106 more-careful responses should therefore be interpreted as a self-reported response pattern, not direct behavioral evidence of verification.

This caution is consistent with Peters, Biermeier, and Scharlau (2026), who found that experimentally measured attention changes after AI errors did not necessarily improve subsequent judgments.

Attention category	n	Composite mean	Terminal scenario mean
Much less carefully / skimming	29	5.579	6.276
Slightly less carefully	19	4.632	5.842
No change	27	4.259	4.167
More carefully	106	4.491	5.877

Inferential Results for H4 and Scale Sensitivity

H4: Frustration did not predict grouped switch-away intentions

Predictor	OR / point	95% CI	p	McFadden R2
frustration composite	1.077	[0.837, 1.384]	0.5645	0.001
A8	0.908	[0.756, 1.090]	0.2997	0.005
B5of10	1.042	[0.858, 1.266]	0.6781	0.001
terminal scenario mean	0.963	[0.790, 1.175]	0.7116	0.001
all scenario mean	1.130	[0.910, 1.404]	0.2694	0.005

The primary composite model gave OR = 1.077, 95% CI [0.837, 1.384], $p = 0.565$. No tested frustration operationalization predicted the grouped switch-away outcome. H4 was not supported under the present operationalization. This does not prove a zero relationship; it means the study did not provide evidence for the proposed grouped association.

Action-category sensitivity

The five first-action categories differed overall, Kruskal-Wallis $H(4) = 32.73$, $p < .001$. The composite mean was highest for giving up (6.000) and lowest for switching to a different AI (3.667). External search, tool switching, and abandonment therefore should not be treated as interchangeable outcomes.

First action	n	Composite mean	Terminal mean
Rephrase or simplify my question	69	4.586	5.696
Switch to Google or another search engine	71	4.693	5.585
Switch to a different AI	21	3.667	4.976
Retry with the same question	4	4.550	6.375
Give up on the task	16	6.000	6.812

Scale reliability and overlap

Cronbach's alpha was 0.694. This is best described as marginal internal consistency for an exploratory, multidimensional composite, not scale validation. The composite contains a giving-up item while H4 includes giving up in the outcome, and its mental-effort item overlaps the attention mechanism proposed in H3. These forms of criterion overlap further support reporting the scenario-frustration sensitivity models.

Sources added for the statistical revision: American Psychological Association, JARS-Quant Table 1, <https://apastyle.apa.org/jars/quant-table-1.pdf>; American Association for Public Opinion Research, Disclosure Standards, <https://aapor.org/standards-and-ethics/disclosure-standards/>; Peters, T. M., Biermeier, K., and Scharlau, I. (2026), <https://doi.org/10.3389/fpsyg.2025.1694367>; Kim, J. et al. (2023), <https://doi.org/10.1145/3557889>; McNeish, D. (2018), <https://doi.org/10.1037/met000144>.

Discussion

Main Finding

The rebuilt analyses confirmed a strong increase in reported frustration across repeated-error checkpoints. They also showed that the proposed verification shift should be treated as an interpretation rather than an observed behavioral mechanism. Many participants reported paying closer attention, rephrasing requests, or selecting Google/search as an intended next action, but the survey did not directly observe verification behavior. The evidence supports a reported verification-oriented response pattern, not a demonstrated within-person change from trusting to verifying (Lee & See, 2004).

Interpretation of H1: Repeated Errors Increased Frustration

H1 was supported. According to the mean frustration levels, it can be noted that it increased as mistakes increased in both scenarios. Scenario A demonstrated a stable upward trend in terms of frustration from the first incorrect response up to the eighth, while Scenario B showed the increase of this variable from early to near-end stages. This pattern is consistent with HCI research showing that repeated errors can interfere with goals and require extra recovery actions (Ceaparu et al., 2004; Lazar et al., 2006). Still, according to the results of

Scenario A, it should be noted that frustration did not grow at an equal rate for all checkpoints.

The biggest increase was noted between the first and third incorrect responses, while the next increase became lower between the fifth and eighth errors. In such a way, the frustration starts growing rather fast after the first mistakes and then stops growing and even starts leveling off after users start expecting the errors from the bot. This could be explained by the loss of trust in the bot as more errors occur.

Interpretation of H2: Evidence Was Mixed

H2 received mixed evidence. In the forced-choice item, 60 participants selected Scenario B and 35 selected Scenario A; among these exclusive choices, exact binomial $p = .013$. However, more participants selected both equally (75) than either scenario alone.

The paired final-checkpoint comparison pointed slightly in the opposite direction: Scenario B near the end was 0.182 points lower than Scenario A after eight wrong answers, Wilcoxon $p = .0095$, $d_z = -0.174$. Scenario B was rated higher by 28 respondents, equal by 101, and lower by 52. Therefore, the forced-choice and paired-rating operationalizations did not converge. Scenario A and Scenario B also differed in total errors, error rate, timing, wording, and fixed order, so this survey cannot isolate inconsistency as the causal factor. H2 should be interpreted as mixed rather than supported.

Interpretation of H3: Any Attention Change, Not a Confirmed Verification Direction

H3 was partially supported. Scenario-frustration measures predicted whether respondents selected any attention-change category, but the five-item composite did not reach the conventional .05 threshold, OR = 1.39, 95% CI [0.97, 1.98], $p = .069$. Directional analysis did not support a simple shift toward more careful reading: higher composite frustration was associated with less-careful categories, Spearman $\rho = -0.228$, $p = .002$. The attention item was also asymmetric and self-reported. The results therefore support an association with reported attention change, but they do not establish that frustration produced more careful verification. Direct behavioral measures would be needed to test that mechanism.

Interpretation of H4: The Grouped Switch-Away Hypothesis Was Not Supported

H4 was not supported under the present operationalization. Logistic regression indicated that the frustration composite did not predict the grouped switch-away outcome, OR = 1.08, 95% CI [0.84, 1.38], $p = .565$. The result should not be interpreted as proof of no relationship; the study did not provide evidence for the proposed grouped association.

External search, switching to another AI, and giving up may represent different processes. Grouping them together may have obscured category-level differences, and future research should model repair, external verification, tool substitution, and abandonment separately.

The Verification-Shift Interpretation

This paper proposes a verification-shift interpretation. The survey findings suggest that repeated chatbot errors may be associated with verification-oriented intentions, including closer reading, rephrasing, citation searching, Google searching, and changing AI tools. However, these were hypothetical self-reports collected after fixed scenarios, not observed within-person behavioral changes. The verification shift should therefore be presented as a conceptual interpretation and a hypothesis for future live-task research rather than as a mechanism demonstrated by the present study. LLM hallucinations remain relevant to why users may perceive a need for verification (Huang et al., 2023; Ji et al., 2023).

Role of Expectations

According to survey data, pulling from the written responses, users' expectations may influence the level of frustration experienced towards the chatbot. Some participants reported that they feel less frustrated with AI due to the expectation of the system's inaccuracy or incompleteness and its necessity to verify facts. Other participants reported feeling more frustrated with the system due to the confidence of AI, neglecting the instructions, repetition of the mistake, and reluctance to adjust the strategy despite corrections. This finding means that frustration is caused not only by the inaccuracy of AI, but also by the breach of users' expectations about usefulness, transparency, and control of AI.

Implications

These findings mean that uncertainty should be communicated, sourcing should be offered, and overconfident answers should be avoided where possible. Implication is consistent with automation-trust literature according to which useful systems should help users form appropriate trust towards it: neither blind dependence nor total mistrust (Parasuraman & Riley, 1997; Lee &

See, 2004). For the perspective of users and students, these results imply that verification skills are as important as prompt-writing skills. For the perspective of researchers, these results imply that frustration should be researched not only emotionally, but also behaviorally.

Limitations

- Since the scenario-based survey relied on imagined chatbot experiences instead of an actual chatbot task, the data represent hypothetical behaviors rather than observed behaviors.

- Attention was self-reported instead of measured using reading time or eye tracking.
- Dropout was assessed using intended action or probable first action and not actual task abandonment.
- The sample size used was a convenient sample and therefore does not represent all AI users.

- The chatbot platform item allowed more than one selection, so platform percentages describe multiple responses and can sum above 100%.

- There was an inconsistency in the error rates between Scenario A and Scenario B; scenario A presented that all answers were errors whereas Scenario B presents both errors and correct answers.
- Duplicate content responses were also found hence the need to indicate if the final claim is based on the raw data or cleaned data.

Future Research

The next step researchers should take, branching off from this current research, is to conduct a similar experiment in a simulated environment of live chatbot, where various aspects can be measured such as the time of reading, the length of prompting, the number of retries, fact-checking activities, and actual dropout. Future research may also include comparing various AI platforms, studying the effect of advanced AI users on tolerance of errors and whether citing or placing uncertainty labels may decrease frustration caused by mistakes.

Additional Limitations and Reporting Notes

The following limitations are added to the original list without removing the limitations already reported in the manuscript.

Fixed order and scenario confounding. All respondents completed the scenarios and checkpoints in the same order. Scenario A and Scenario B also differed in total errors, error rate, timing, consistency, and wording. H2 therefore does not isolate inconsistency as a causal factor.

Self-report rather than observed behavior. The survey measured imagined frustration, attention, and first-action intentions. It did not observe reading time, source checks, reliance, switching, or abandonment in a live chatbot task.

Asymmetric attention item. The response options included one broad more-carefully option and two levels of less-careful reading. This limits directional interpretation of the attention frequencies.

Response-content patterns are not verified identities. The 181 retained records are unique answer patterns under the exact-content rule. The data did not include respondent identifiers, device/IP markers, or completion-time paradata sufficient to verify 181 unique people.

Post hoc cleaning documentation. The exact-duplicate rule is reproducible, but the supplied materials did not contain a preregistered exclusion plan, attention checks, or a frozen exclusion log created before analysis.

Nonprobability sample. The convenience sample cannot support population prevalence estimates for all AI users. Data were collected from April 5 through May 31, 2026. Platform selections allowed multiple responses, so the platform percentages can sum above 100%.

Ethics reporting. Twenty-four retained records selected Under 18. The supplied manuscript does not document the applicable consent, assent, parental-permission, or institutional/school ethics determination. The relevant procedure should be added if available before the paper is submitted or distributed as completed human-subjects research.

Exploratory composite. The five-item composite combines affect, effort, confidence, time appraisal, and giving-up intention. Alpha alone does not establish a single validated construct, and some item content overlaps the H3/H4 outcomes.

Illustrative written responses. The 37 optional comments were not accompanied by a prespecified codebook, coder information, reflexivity statement, or formal theme-frequency analysis. They are therefore presented as illustrative comments rather than a formal thematic analysis.

Conclusion

This study aimed to investigate the relationship between the accumulation of errors committed by AI chatbots and frustration, trust, attention, and intended user behavior. Rebuilt analyses strongly supported H1: frustration ratings increased across both hypothetical error sequences. Evidence for H2 was mixed because the forced-choice and paired-rating comparisons did not agree. H3 was partially supported for any reported attention change, but the study did not establish a directional shift toward more careful verification. H4 was not supported because frustration did not predict the grouped switch-away outcome. Many respondents reported closer reading, rephrasing, and Google/search as intended responses. These patterns support treating verification-oriented behavior as a promising interpretation for future live-task research, not as a causal mechanism demonstrated by this scenario survey. The study contributes exploratory evidence that repeated perceived AI errors are related to frustration and to several distinct reported adaptation intentions.

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Appendix A: Survey Instrument Summary

This survey covered users usages of chatbot platforms, frequency of uses, AI experiences, perceived accuracy of information provided, overall trustworthiness, reduced trustworthiness from answering incorrectly, level of frustration in Scenario A, level of frustration in Scenario B, failure-type comparison, shift in attention to reading, attention shift threshold, first response to

repeatedly incorrect answers, a five-item frustration scale, overall reliability of chatbot, overall predictability of chatbot, optional explanations and demographics.

Appendix B: AI Use Transparency Statement

AI programs have been used for creating outlines, organizing thoughts, drafting, preparing charts, explaining statistics, and assisting with the rebuilt statistical models and revision formatting. The student researcher remains responsible for examining all material, validating the data, reviewing the model specifications, rewriting or approving the paper language, and ensuring the accuracy of statements based on the survey results.